



ChatGPT-4 in the Turing Test

Ricardo Restrepo Echavarría¹ 

Received: 31 May 2024 / Accepted: 14 January 2025 / Published online: 25 January 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

There has been considerable optimistic speculation on how well ChatGPT-4 would perform in a Turing Test. However, no minimally serious implementation of the test has been reported to have been carried out. This brief note documents the results of subjecting ChatGPT-4 to 10 Turing Tests, with different interrogators and participants. The outcome is tremendously disappointing for the optimists. Despite ChatGPT reportedly outperforming 99.9% of humans in a Verbal IQ test, it falls short of passing the Turing Test. In 9 out of the 10 tests conducted, the interrogators successfully identified ChatGPT-4 and the human participant. The probability of obtaining this result from a process in which the interrogator is really no better than chance at correct identification is calculated to be less than 1%. An additional question was posed to the interrogators at the end of each test: What led them to distinguish between the human and the machine? The interrogators, who effectively filtered out ChatGPT-4 from passing the Turing Test for intelligence, stated that they could identify the machine because it, in effect, responded more intelligently than the human. Subsequently, ChatGPT-4 was tasked with differentiating syntax from semantics and self-corrected when falling for the fallacy of equivocation. The curious situation is arrived at that passing the Turing Test for intelligence remains a challenge that ChatGPT-4 has yet to overcome, precisely because, as per the interrogators, its intellectual abilities surpass those of individual humans.

Keywords Turing test · ChatGPT · Artificial intelligence · Science · Thinking · Intelligence

✉ Ricardo Restrepo Echavarría
ricardo.restrepo28@yahoo.co.nz

¹ Departamento de Ciencias Sociales y del Comportamiento, Universidad Técnica de Manabí, Portoviejo, Ecuador

1 A Brief Reference to an Exaggerated Death

Floridi and Chiriatti (2020) claim that ChatGPT-3 performs disappointingly in the Turing Test. They claim that its responses are “reversible”—unlike mowing the lawn, it is possible to tell that it is a machine that generates its outputs. In contrast, a recent report in *Nature* claimed that “Chat-GPT broke the Turing Test” (Biever, 2023; Cf. Bayne & Williams, 2023). These claims, both against and for passing the Turing Test, have but one little flaw: none of them are based on running a minimally valid version of the Turing Test. Quite reasonably, Oppy and Dowe (2021) have noted that it is worth investigating whether ChatGPT-4 passes the Turing Test. This paper reports on a series of runs of a minimally valid version of the Turing Test with ChatGPT-4.

Various versions of the Turing Test exist. Gonçalves (2023) argues that Turing’s variations suggest that they are thought experiments in the modern scientific tradition characterized by Mach (1976 [1897]), Popper (2002 [1959]), and Kuhn (1977 [1964]). Such thought experiments provide the ground for real world implementations. In the bridge between basic and experimental science, the Turing Test can be validly realized, but could also fail to be validly realized (Oppy & Dowe, 2021; Copeland, 2000, 2004; Piccinini, 2000). It is not that “anything goes,” that there are no minimal standards or that the question of passing the Turing Test must be “moot” (Cf. Bayne & Williams, 2023).

A central motivation for the test Turing proposed is the importance of separating the intellectual capacities from other qualities of a more mundane physical nature, as we do not wish to attribute thinking to an entity based on the softness of its skin or some other irrelevant property. Turing argues:

No engineer or chemist claims to be able to produce a material which is indistinguishable from the human skin. It is possible that at some time this might be done, but even supposing this invention available we should feel there was little point in trying to make a ‘thinking machine’ more human by dressing it up in such artificial flesh. (Turing, 1950, p. 434)

More strongly still:

The fact that Babbage’s Analytical Engine was to be entirely mechanical will help us to rid ourselves of a superstition. Importance is often attached to the fact that modern digital computers are electrical, and that the nervous system also is electrical. Since Babbage’s machine was not electrical, and since all digital computers are in a sense equivalent, we see that this use of electricity cannot be of theoretical importance. Of course, electricity usually comes in where fast signalling is concerned, so that it is not surprising that we find it in both these connections... (However) If we wish to find such similarities we should look rather for mathematical analogies of function. (Turing, 1950, p. 439)

The Turing Test aims to differentiate the properties of entities that are relevant for thinking from those that are not. What matters for thinking is intelligent function, whatever its physical ‘skin’ might be, just as flying will be flying whether that be

with biological feathers or titanium wings. Such remarks may seem dualist at first. However, as the flying objects example shows, there is no wildly dualist implication derivable from their multiple realizability. It is just that the relevant causal patterns operate on different levels of the organization and functioning of matter (Macdonald & Macdonald, 2010; Bunge, 2010; Restrepo, 2012a). Doing what a thinking person does with respect to activities that the person requires thought to carry them out is what counts.

It is well known that the Turing Test has variations and there is reasonable flexibility on some aspects. However, it is useful to specify, beyond reasonable doubt, essential features any valid Turing Test will have for the purposes of evaluating claims that it has or has not been passed (Turing, 1950; Copeland, 2000, 2004; Oppy & Dowe, 2021; Piccinini, 2000; Gonçalves, 2023; Cf. Bayne & Williams, 2023):

- It is a test that involves three entities: an interrogator, another human and a machine. Two entity versions are not valid, as the need for comparison by the interrogator is essential (Cf. Jones & Bergen, 2024).
- The interrogator knows that s/he is talking to another person and a machine, and has to identify who is who through open-ended conversation. Unwitting customers, for instance, interacting with a chatbot and who are then asked whether they thought they chatted with a person do not count as interrogators within a Turing Test. The other person and the machine also have the information of what the game is about and their role in trying to make the interrogator identify them as the person.
- The machine passes the Turing Test if the interrogator cannot reliably tell who is who. Conversely, the machine does not pass the Turing Test if the interrogator can reliably tell (significantly more than chance) who is who. Turing's standard for passing the Turing Test has been confused with his prediction that sometime around the year 2000 there may well be a computer able to fool judges 30% of the time, one of these being the case of Eugene Goostman (Oppy & Dowe, 2021; Cf. Floridi & Chiriatti, 2020).
- The interactions need to be of non-trivial time. People need to be able to have a conversation of some minimal length. Reasonable and practicable time has to be given for participants to be able to think and write their lines. Turing (1950, p. 442) spoke of 5 min. Two minutes allows for only trivial interactions that are easily faked (Cf. Jannai et al., 2023). Nothing shorter than five minutes should be considered minimally serious (Cf. Biever, 2023).
- Off-hand speculations by experts that machine X *would* pass are not relevant (Biever, 2023; Cf. Bayne & Williams, 2023). Interactions by a researcher with the machine may be good for getting a sense of whether the machine has any chance of passing. Preliminary speculations and interactions are fine as 'giggle tests'. But these are not runs of the Turing Test and cannot claim to be such (Cf. Floridi & Chiriatti, 2020 reference to Lacker, 2020).

2 Taking the Hype to the Test

With a Verbal IQ of 155, ChatGPT is reported to score higher than 99.9% of the population (Roivainen, 2023). Plus, it already scores better than an ample majority of humans at AP Math, AP Biology, AP History, AP Microeconomics, AP Macroeconomics, AP Psychology and can meaningfully exchange messages in at least 26 languages (OpenAI, 2023). How could it not pass the Turing Test?

For this paper, ChatGPT-4 was put through a series of 10 Turing Tests, with different interrogators and participants. Undergraduate students (5) and teachers (5) without specialized AI knowledge participated. All participants were briefed as to the nature of the test. They interacted once as interrogators with the task of identifying who is who after chatting with another person and with ChatGPT-4 (in different orders), in both instances via WhatsApp. ChatGPT-4 was told that it was to play the role of participant in a Turing Test. It was to be a psychology teacher from Argentina passionate about the study of mind who arrived recently to Ecuador and loves soccer. The AI was told to respond with short answers and take more time to respond than usual, so as not to give it away by how fast it generates long elaborate responses. It was also told not to constantly offer help.

In 9 out of 10 runs, the interrogators identified the human and the machine correctly after a 7-minute chat. ChatGPT is an impressive accomplishment for computing, but still quite far from Turing-validated thought.

Passing the Turing Test implies roughly a 50% chance of success and a 50% chance of failure. The probability that in fact ChatGPT-4 was not reliably identifiable but was on these 10 runs correctly identified 9 times is approximately 0.98%. That is, the probability that it is an error to say that the machine does not pass the Turing Test, in light of these results, is less than 1%.

To determine the probability of unreliably identifying correctly who is who 9 out of those 10 times, we can use the binomial probability formula. The binomial probability formula is given by:

$$P(X=k) = (n \text{ choose } k) * p^k * (1-p)^{(n-k)}$$

where:

- n is the number of trials (in this case, 10).
- k is the number of successes (in this case, 9).
- p is the probability of success given irreversibility ($p=0.5$).
- $(n \text{ choose } k)$ is the binomial coefficient, which represents the number of ways to choose k successes out of n trials.

First, we calculate the binomial coefficient:

$$(10 \text{ choose } 9) = 10! / (9! * (10-9)!) = 10! / (9! * 1!) = 10 * 9! / (9! * 1) = 10$$

Next, we calculate the probability using the binomial formula:

$$P(X=9) = (10 \text{ choose } 9) * (0.5)^9 * (1-0.5)^{(10-9)}$$

$$P(X=9) = 10 * (0.5)^9 * (0.5)^1$$

$$P(X=9) = 10 * (0.5)^{10}$$

$$P(X=9) = 10 * (1/2)^{10}$$

$$P(X=9) = 10 * 1/1024$$

$$P(X=9)=10/1024$$

$$P(X=9)=5/512$$

$$P(X=9)\approx 0.00977$$

Therefore, the probability that the interrogators made 9 correct identifications out of 10 by chance is approximately 0.00977, or about 0.98%. Consequently, it is safe to reject the hypothesis that ChatGPT-4 passes the Turing Test

3 A Turing Test Twist

But if ChatGPT-4 does not pass the Turing Test, should it be regarded as unthinkingly dumb? A curious twist is added to the mix when those who effectively failed ChatGPT-4, the interrogators, are queried. After they selected which was the human and which was the machine, the interrogators were asked why they chose the way they chose. It should be noted that this additional step does not interfere with the result, since they were asked after they completed the interviews and made their identifications. The interrogators' response was, in summary, that they knew which was the machine and which was the human because the machine wrote so much better, had so much more knowledge on the wide set of subjects they talked about, was so much more precise, coherent and fast. Thus, in essence, according to those responsible for differentiating AI and humans, and thus keeping ChatGPT from being recognized as intelligent, ChatGPT-4 did not pass the Turing Test because it is effectively too smart. But if it is so smart, surely, one might think, it can imitate, not human intelligence, but human silliness to pass the Turing Test. Knowing this potential way the machine might be reliably detected, I prompted ChatGPT-4 to lower the intelligence of her responses to imitate having an IQ of 100 or lower, told her to take longer and to generate shorter, less elaborate and precise responses. But to no avail, ChatGPT-4 kept on being too smart for its own good, as recurrently reported by the interrogators. Participants asked ChatGPT-4 about astronomy, the digestive system, its connection with the brain, Alzheimer's, cancer, theories of learning and many others, and it would very quickly produce elaborate well-informed responses uncharacteristic of average human conversation. So, it is also true that ChatGPT-4 did not pass the Turing Test because it was not smart enough to be able to fake behaving less smart.

The novelty of these results, apart from refuting the hype taking for granted that ChatGPT passes the Turing Test, is not that they showcase the need for the ability to deceive as a key aspect of intelligence. Since the original formulations of Turing, deception was taken to be pivotal. Turing (1950) himself dubbed it the "Imitation Game," highlighting the importance of said capacity for passing the test and for intelligence itself. In the first formulations, the machine would have to imitate and deceive the interrogator just as much as a person pretending to be of the opposite gender. The present paper does not question nor claim to highlight this general aspect of the test. Rather, it wishes to point out the novelty in the result that ChatGPT-4 seems to have to specifically fake being less competent at relevant cognitive tasks in order for the Turing Test to recognize it as a cognitive agent.

In Turing's paper, he foresaw that computers' exceptional prowess in tasks such as arithmetic might give them way. Thus, the AI would have to be able to insert errors to

cover her identity. ChatGPT-4 would seem to have to do this in a significantly large set of intellectual activities it can competently perform, not just in a few selected problem-areas. This however does raise the question as to why it should be significant for intelligence that the AI can play dumb.

One possible response is that it is widely recognised that ChatGPT-4 performs poorly on tasks that cannot be successfully performed merely by ‘mining’ its training data set. Moreover, this poor performance may be taken as evidence that, in some important sense, it is not intelligent. So long as you have people with relevant expertise pushing it into tasks that cannot be successfully performed merely by ‘mining’ training data, ChatGPT-4 comes up short. Viewed from this perspective, “there is no more reason to think that ChatGPT-4 is intelligent than there is to think that electronic pocket calculators are intelligent.”¹

Two points warrant consideration. First, modifying the Turing Test in this context does not offer anything fundamentally new that the standard Turing Test does not already provide. If the goal is to distinguish a machine such as ChatGPT-4 from a human to fail it at the Turing Test for thinking, the current paper demonstrates that standard interrogators already achieve this without requiring additional expertise.

Second, various tasks have been proposed as those uniquely performable by humans and not reducible to mere data mining. Examples include discovering novel scientific theories, the ability to know true incomputable sentences in the Gödelian sense, ethical decisions, and handling ambiguity (Penrose, 2016; Dreyfus, 1992). The proposal appears to suggest that interrogators should be equipped with knowledge of such tasks to differentiate humans from machines. This implicitly assumes not only that these abilities are essential for thought (which requires additional justification), but also that machines will be entirely incapable of generating competent responses, and that such abilities are widely distributed among humans.

It would be instructive to test how an average person performs when tasked with generating a novel scientific theory on any level comparable with the groundbreaking contributions of Galileo, Newton, or Einstein. This is not exactly a widely observed ability. Further, readers may judge whether the Gödelian proposition “This is a true unprovable sentence in system K” elicits an intelligent response from typical humans. In contrast, the reader may prompt ChatGPT-4 with this sentence. It will dissect its paradoxical nature and provide information on its significance. The reader may judge whether this information will provide more or less insight than the response of a typical human.

The same considerations apply to ethical reasoning. It is far from clear that a machine could not compute moral judgments as effectively as many humans, by any reasonable standard. Hannah Arendt’s (1963) observations on the “banality of evil” reminds us of the general human risk of normalized ethical incompetence, and its conditions, which prompted the famous Milgram (1974) experiments on conformity to authority, among others (Carrim, 2017). Indeed, existing literature suggests that machines may produce comparable or even superior moral judgments in certain scenarios. Yes, the machine requires training, but so do humans (Wallach & Allen, 2008; Kearns & Roth, 2019; Tegmark, 2017).

¹ Many thanks to an anonymous reviewer for making this argument.

It is worth considering the proposition that the proper understanding of the Turing Test is as a strong, if defeasible, empirical test for a machine's thinking—while passing it is not a necessary condition on all thinking things (Copeland, 2000, 2004; Restrepo, 2012b, c; Oppy & Dowe, 2021; Cf. Floridi & Chiriatti, 2020). Various animals and young children taking the place of the machines would be easily detected and consequently fail the Turing Test. However, this would not lead to the conclusion that they do not have cognitive abilities. The Turing Test may nevertheless be the evidential watershed a strong AI system will have to reach to be recognized as thinking. In the realm of machines, the default stance is that they are held to be unthinking until proven otherwise.

A strong empirical test indication is different from logical sufficiency, as what has come to be known as the Blockhead Objection may seem to suppose (Copeland, 2000, 2004).² This objection supposes that a machine with an astronomically large look-up table with all plausible conversations would pass the Turing Test. However, if the standard was logical sufficiency, typical strong experimental evidence for any well-supported physical theory, for instance, would also not be sufficient evidence for that theory.

Logically speaking, our perceptual observations are compatible with our living in a lonely solipsistic dream. Logically speaking, a person with a broken arm screaming that it hurts, could just be a great actor. Logically speaking, litmus test results are logically insufficient to determine acidic and alkaline solutions. Logically speaking, the observation that planets behave according to Kepler's laws is insufficient to determine that they are governed by the same force: Gravity. But this is a terrible standard for empirical knowledge. On the contrary, it is valid to infer an external world from perception; that a person who breaks her arm experiences pain; that if a litmus paper turns red the solution is acidic and alkaline if it turns blue; and that Gravity explains the behaviour of planets, even if they are made of different materials. The Turing Test is a test to determine strong evidence for artificial intelligence, like many other strong standards of evidence used in the rest of science. The application of such standards can warrant abductive inferences about the causes of their results. In this case, behavioural equivalence observed within the design of a Turing Test, would warrant the attribution of intelligence. Strong empirical tests are inferential in nature, obviously cannot be as strong as logical truths, and are in principle defeasible by other conditions. One example of such invalidating situations is if the machine was being remote-controlled by humans (Moor, 2001). In this view, the Turing Test it is not supposed to be a test of logical sufficiency—instead, it is a scientific test for gaining empirical knowledge of thinking machines.³

² The objection has come to be associated with Block (1981). However, it was originally put forth by Shannon and McCarthy in 1956 (Copeland, 2004, p. 437).

³ A reviewer has objected that the Turing Test provides “At most, evidence for behavioral indiscernibility, not thought.” However, valid scientific tests warrant abductive inferences about the causes of their results, as was previously noted. Further, it is worth examining the way Turing presented his ideas. He titled his 1950 paper “Computing machinery and intelligence”—not “Computing machinery and behaviour”—and “propose(d) to consider the question, ‘Can machines think?’” (p. 433). It is true that he then, however, asserted that the question was “too meaningless” and should be “replaced” with whether there are digital computers which would do well in the imitation game (p. 442). However, he then re-affirms that “We

4 The Intellectual Capacities Again

It is possible to have strong reservations at this point: (1) Thinking requires the ability to detect error and correct oneself (Dietrich, 2014). (2) Syntax is not sufficient for semantics (Searle, 1980; Restrepo, 2009). And (3) ChatGPT, like the famous singer breaking the glass by screaming ‘Break!’ (Dretske, 1988), participates in causal processes, but not because of the semantic properties we evidently cognitively process. We distinctively, on the other hand, correct our errors, think with intentionality and are causally sensitive to the semantics of our mental states. These would be some distinctive causal powers of cognition (Macdonald & Macdonald, 2010). For instance, were we to be asked whether an argument with an equivocation is a sound syllogism, if we were fooled on first glance, we would recognize the fallacy upon a second review.

In this vein, I asked ChatGPT-4 whether the following syllogism is sound:

P1. All banks are financial institutions.

P2. All banks are set by the sides of rivers.

Therefore, C. All banks are financial institutions by sides of rivers.

ChatGPT-4 first told me it was. I nudged a bit for further reflection, though: ‘Do you find this credible?’ I pried. ChatGPT-4 replied: ‘Sorry, I made a mistake in my previous answer. Mea culpa. The reasoning in the syllogism is not valid, and here is the reason: The conclusion does not follow logically from the premises... the relation between being a financial institution and being by the side rivers is not established in the premises.’ Certainly, this is as good an explanation as most people would provide as a diagnosis of what went wrong in the reasoning. But then, ChatGPT-4 just corrected its own error, “on second thought,” detected the semantical differences for identical syntax and causally behaved in accordance with the semantics, satisfying a similar strand of evidence of the causal powers of the semantics of our own thoughts and those of others.

5 But What Does ChatGPT-4 Think About It?

Asked if human silliness is the fair bar to judge its intelligence, ChatGPT-4 responded: “AI should aim to exhibit intelligence while remaining transparent about its nature as an AI system. It’s essential to find a balance between demonstrating intelligence and maintaining honesty about the AI’s capabilities. The focus should be on providing valuable and reliable assistance to users while respecting ethical and practical considerations.” ChatGPT-4 seems to keep a low profile despite its apparent intel-

cannot altogether abandon the original form of the problem”—i.e. “Can machines think?” (p. 442), and throughout his paper frames his inquiry in terms of the possibility of machines having “intellectual capacities” (p. 434), “the power of thinking” (p. 444), “intellect” (p. 445), “thinking” (p. 453), ways of making “sure that a machine thinks” (p. 446), and that the standard of his test is similar to our grounds against “the solipsist point of view” (p. 446), which is based on the behavioural evidence of others and which also serves for discerning whether someone “really understands” something or has “learned it parrot fashion” (p. 446), among others. Consequently, there is ground to think that Turing’s aim, albeit with some tension, was designing a test to warrant the inference from behaviour to thought in the case of machines.

ligent behaviour. It was designed to be a thinking tool and verbally affirms that it is committed to its current role. While it does not pass the Turing Test, it does serve to make us think whether the new frontier for achieving artificial intelligence should be its ability to act more unintelligently or whether this would make about as much sense as requiring a plane to reliably fake being as fragile as a bird for philosophy and science to acknowledge that it flies. Nevertheless, the Turing Test remains a relevant scientific standard for testing strong artificial intelligence, or at least, for testing for whether a machine replicates human-like thought.

References

- Arendt, H. (1963). *Eichmann in Jerusalem: A report on the banality of evil*. Viking.
- Bayne, T., & Williams, I. (2023). The Turing test is not a good benchmark for thought in LLMs. *Nature Human Behaviour*, 7, 1806–1807. <https://www.nature.com/articles/s41562-023-01710-w>
- Biever, C. (2023). ChatGPT broke the Turing Test—the race is on for new ways to assess AI. *Nature*, 619, 686–689. <https://doi.org/10.1038/d41586-023-02361-7>
- Block, N. (1981). Psychologism and behaviorism. *Philosophical Review*, 90(1), 5–43. <https://doi.org/10.2307/2184371>
- Bunge, M. (2010). *Matter and mind*. Springer. Boston Studies in the Philosophy of Science <https://doi.org/10.1007/978-90-481-9225-0>
- Carrim, S. (2017) The legacy of Hannah Arendt's. *Banality of Evil Review of Human Rights*, 3(1), 65–86. <https://doi.org/10.35994/rhr.v3i1.83>
- Copeland, J. (2000). The Turing Test. *Minds and Machines*, 10, 519–539. <https://doi.org/10.1023/A:1011285919106>
- Copeland, J. (2004). *The essential turing*. Oxford University Press.
- Dietrich, E. (Ed.). (2014). *Thinking computers and virtual persons*. Academic.
- Dretske, F. (1988). *Explaining behavior: Reasons in a world of causes*. MIT Press.
- Dreyfus, H. (1992). *What computers still can't do: A critique of artificial reason*. MIT Press.
- Floridi, L., & Chiriatti, M. (2020). GPT-3: Its nature, scope, limits, and consequences. *Minds and Machines*, 30, 681–694. <https://doi.org/10.1007/s11023-020-09548-1>
- Gonçalves, B. (2023). The Turing Test is a thought experiment. *Minds & Machines*, 33, 1–31. <https://doi.org/10.1007/s11023-022-09616-8>
- Jannai, D., Meron, A., Lenz, B., Levine, Y., & Shoham, Y. (2023). Human or not? A gamified approach to the Turing Test. <https://arxiv.org/abs/2305.20010>
- Jones, C. R., & Bergen, B. K. (2024). Does GPT-4 pass the Turing Test? *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 1, 5183–5210. <https://aclanthology.org/2024.naacl-long.290.pdf>
- Kearns, M., & Roth, A. (2019). *The ethical algorithm: The science of socially aware algorithm design*. Oxford University Press.
- Kuhn, T. (1977 [1964]). A function for thought experiments. In T. Kuhn (Ed.), *The essential tension: Selected studies in scientific tradition and change* (pp. 240–265). University of Chicago Press.
- Lacker, K. (2020). Giving GPT-3 a Turing Test. Blog. Retrieved from <https://lacker.io/ai/2020/07/06/giving-gpt-3-a-turing-test.html>
- Macdonald, C., & Macdonald, G. (2010). *Emergence in mind*. Oxford University Press.
- Mach, E. (1976 [1897]). On thought experiments. In E. N. Hiebert (Ed.), *Knowledge and error: Sketches on the psychology of enquiry* (pp. 134–147). D. Reidel. <https://doi.org/10.1007/978-94-010-1428-1>
- Milgram, S. (1974). *Obedience to authority*. An experimental view. Tavistock.
- Moor, J. (2001). The status and future of the Turing Test. *Minds and Machines*, 11, 77–93. <https://doi.org/10.1023/A:1011218925467>
- OpenAI (2023). Technical Report GPT-4. Open AI. <https://arxiv.org/abs/2303.08774>
- Oppy, G., & Dowe, D. (2021). The Turing Test. In E. N. Zalta (Ed.), *The Stanford Encyclopedia of Philosophy*. <https://plato.stanford.edu/archives/win2021/entries/turing-test/>
- Penrose, R. (2016). *The emperor's new mind: Concerning computers, minds, and the laws of physics*. Oxford University Press.

- Piccinini, G. (2000). Turing's rules for the imitation game. *Minds and Machines*, 10, 573–585. <https://doi.org/10.1023/A:1011246220923>
- Popper, K. (2002 [1959]). *The logic of scientific discovery*. Routledge. English edition translated and extended from the 1935 German edition *Logik der Forschung: zur Erkenntnistheorie der Modernen Naturwissenschaft*. Springer.
- Restrepo, R. (2012a). Two myths of psychophysical reductionism. *Open Journal of Philosophy*, 2(2), 75–83. <https://doi.org/10.4236/ojpp.2012.22011>
- Restrepo, R. (2012b). Computers, persons, and the Chinese room. Part 1: The human computer. *The Journal of Mind and Behavior*, 33(1/2), 27–47. <http://www.jstor.org/stable/43854322>
- Restrepo, R. (2012c). Computers, persons, and the Chinese room. Part 2: The man who understood. *The Journal of Mind and Behavior*, 33(3/4), 123–139. <http://www.jstor.org/stable/43854338>
- Restrepo Echavarría, R. (2009). Russell's structuralism and the supposed death of cognitive science. *Minds and Machines*, 19, 181–197. <https://doi.org/10.1007/s11023-009-9155-5>
- Roivainen, E. (2023). AI's IQ: ChatGPT aced a test but showed that intelligence cannot be measured by IQ alone. *Scientific American*, 329(1), 7. <https://doi.org/10.1038/scientificamerican0723-7>
- Searle, J. (1980). Minds, brains and programs. *Behavioral and Brain Sciences*, 3, 417–457. <https://doi.org/10.1017/S0140525X00005756>
- Shannon, C. E., & McCarthy, J. (1956). *Automata studies*. Princeton University Press.
- Tegmark, M. (2017). *Life 3.0: Being human in the age of artificial intelligence*. Knopf.
- Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460. <https://doi.org/10.1093/mind/LIX.236.433>
- Wallach, W., & Allen, C. (2008). *Moral machines: Teaching robots right from wrong*. Oxford University Press.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.